Source: Sahil Bhola, Self-contained Proof for a Mixed-precision Block Jacobi Algorithm.

Selected deliverable: the full active result labeled corollary:block_dot_product_sequential in main.tex (lines 312-343).

The original mathematical proof is available in paper.tex. The selected statement below is authoritative. Earlier statements and notation supply context. A referenced result may be proved as a helper. Do not assume a desired algorithmic bound or an unproved paper lemma as an added hypothesis.

SELECTED STATEMENT (verbatim LaTeX, proof omitted)
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\begin{corollary}[Sequential Reduction of the Partial Dot Products]
\label{corollary:block_dot_product_sequential}
Under the assumptions of~\cref{theorem:block_dot_product}, suppose that
the $n^\dagger=\ceil{n/T}$ partial dot products are combined by sequential
summation rather than binary-tree reduction. If
$(T+n^\dagger-1)\urd<1$, then the computed result $\hat{y}$ admits either
of the backward-error representations
\begin{align}
    \hat{y}
    &=\sum_{i=1}^{n}
      \left(\vect{x}_i+\Delta\vect{x}_i\right)\vect{y}_i,
    &
    \abs{\Delta\vect{x}_i}
    &\leq \gamma_{T+n^\dagger-1}\abs{\vect{x}_i},
    \\
    \hat{y}
    &=\sum_{i=1}^{n}
      \vect{x}_i\left(\vect{y}_i+\Delta\vect{y}_i\right),
    &
    \abs{\Delta\vect{y}_i}
    &\leq \gamma_{T+n^\dagger-1}\abs{\vect{y}_i},
\end{align}
for all $i=1,\ldots,n$.
\end{corollary}

SECTION CONTEXT
\section{Block Dot Product}
\label{sec:block_dot_product}
Another widely used kernel is computing the dot product kernel
\begin{align}
    y = \sum_{i=1}^n \vect{x}_i \vect{y}_i,
\end{align}
in a block-wise manner given by
\begin{align}
    y = \sum_{j = 1}^{n^\dagger}\sum_{t=1}^{T} \vect{x}_{(j-1)T + t} \vect{y}_{(j-1)T + t} =
\sum_{j=1}^{n^\dagger} p_j, \quad p_j \define \sum_{t=1}^{T} \vect{x}_{(j-1)T + t} \vect{y}_{(j-1)T + t},
\end{align}
where $n^\dagger \define \lceil n / T \rceil$ denotes the number of blocks of size $T$.
Typically, in parallel computing architecture like SIMT, the partial dot-product computation $p_j$ is computed by independent threads and is computed sequentially.
Followed by, to reduce the partials $p_j$, we can perform computations such as sequential summation or parallel binary tree reduction of depth $\log_2 n^\dagger$.
We first perform backward error analysis where the paritals $p_j$ are reduced using parallel binary tree reduction.
Followed by, we also consider the case of sequential accumulation.

[Earlier statement and proof omitted here.]


Compared to the sequential dot product that results in a problem size dependent constant $\gamma_{n}$~\cite{higham2002accuracy}, the improvement due to block-wise computations is given by
\begin{align}
    \kappa_{\mathrm{BDP}} \define \frac{\gamma_n}{\gamma_{T + \lceil \log_2 \lceil n / T \rceil \rceil}},
\end{align}
which can be maximized to find the optimal block size by solving the optimization problem
\begin{align}
    T^\star = \arg\max_T \kappa_{\mathrm{BDP}}.
\end{align}
Since $\gamma_{n}$ is monotonic for $\urd > 0$, we can find the optimal block size by solving a simpler optimization problem given by
\begin{align}
    T^\star = \arg\max_T \frac{n}{T + \lceil \log_2 \lceil n / T \rceil \rceil}.
\end{align}
For a given $n$, ignoring the integer effects, and treating T as continuous, $\kappa_{\mathrm{BDP}}$ is then maximized for $T^\star = \frac{1}{\ln 2}$.
This in practice translates to choosing $T \in \{1, 2\}$.
Since the depth of the binary tree only grows logarithmically, choosing smaller values of the block size $T$ introduces the smaller error accumulation during the sequential computations to obtain $p_j$.
However, in practice, while choosing smaller $T$ is better for backward stability improvements, it is typically not justified from an algorithmic efficiency standpoint.
Interestingly, for $T=1$, we recover the pair-wise dot-product proposed in~\cite[p. 64]{higham2002accuracy}.

EARLIER STATEMENTS FOR CONTEXT
\begin{theorem}[Block Dot Product]\label{theorem:block_dot_product}
Let
\begin{align}
    y=\sum_{i=1}^{n}\vect{x}_i\vect{y}_i
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\end{align}
be evaluated by partitioning the terms into
$n^\dagger=\ceil{n/T}$ blocks of size $T$, computing each block dot
product sequentially, and combining the resulting partial sums using a
binary-tree reduction of depth
$\ell^\dagger=\ceil{\log_2 n^\dagger}$. Any entries beyond $n$ in the
final block are taken to be zero. Assume that no overflow or underflow
occurs and that $(T+\ell^\dagger)\urd<1$. Then the computed result
$\hat{y}$ admits either of the backward-error representations
\begin{align}
    \hat{y}
    &=\sum_{i=1}^{n}
      \left(\vect{x}_i+\Delta\vect{x}_i\right)\vect{y}_i,
    &
    \abs{\Delta\vect{x}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{x}_i},
    \\
    \hat{y}
    &=\sum_{i=1}^{n}
      \vect{x}_i\left(\vect{y}_i+\Delta\vect{y}_i\right),
    &
    \abs{\Delta\vect{y}_i}
    &\leq \gamma_{T+\ell^\dagger}\abs{\vect{y}_i},
\end{align}
for all $i=1,\cdots,n$, where
$\gamma_{T+\ell^\dagger}
=(T+\ell^\dagger)\urd/(1-(T+\ell^\dagger)\urd)$.
\end{theorem}

COMMON ROUNDING MODEL AND ERROR CONSTANTS
\section{Floating-Point Model}
We assume the standard floating-point model~\cite{higham2002accuracy}: for every elementary
operation $\circ \in \{+,-,\times,/\}$,
\begin{equation}
    \fl(a \circ b) = (a \circ b)(1+\delta),
    \qquad
    \abs{\delta} \le \urd,
    \label{eq:fp-model}
\end{equation}
where $\urd$ is the unit roundoff and $\delta$ is the relative roundoff error.
Using~\cref{eq:fp-model} for numerical analysis we typicall obtain the product
\begin{align}
    \prod_{i=1}^n (1 + \delta_i)^{\rho_i} \define 1 + \theta_n, \qquad \rho = \pm 1,
    \label{eq:theta_n}
\end{align}
where $n$ denotes the number of accumulated roundoff error terms.
For numerical analysis, we seek reliable and sharp bounds for $\theta_n$.


\section{Deterministic Rounding Error Analysis}
The deterministic rounding error analysis proposes the following problem-size dependent constant
\begin{align}
\gamma_{n} \define \frac{n\urd}{1 - n\urd},
\label{eq:drea}
\end{align}
for $n\urd < 1$.
Here, $n$ is the number of accumulated error terms~\cite{higham2002accuracy}.
The problem-size dependent constant is monotonic, that is, $\gamma_{n+1} > \gamma_{n}$ for $\urd > 0$.
Performing backward/forward error analysis using~\cref{eq:drea} we obtain
\begin{align}
    \abs{\theta_n} \leq \gamma_n.
\end{align}

SOURCE NOTATION MACROS
\newcommand{\urd}{\ensuremath{\mathrm{u}}}
\newcommand{\define}{\ensuremath{\triangleq}}
\newcommand{\vect}[1]{\mathbf{\MakeLowercase{#1}}}
\newcommand{\mat}[1]{\mathbf{\MakeUppercase{#1}}}
\newcommand{\ireal}[1]{\in\mathbb{R}^{#1}}
\newcommand{\norm}[2][]{\left\|#2\right\|_{#1}}
\DeclareMathOperator{\fl}{fl}
\newcommand{\R}{\mathbb{R}}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\ceil}[1]{\left\lceil #1 \right\rceil}
\newcommand{\widehatgamma}[1]{\widehat{\gamma}_{#1}}
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